CURRENT
FINAL TERM (MCQS)
ZOO501
PREPARE BY: ASHAM+MAAH

TOPIC NO.1

1. _____which divide mitotically to produce all the cells of the body…..… **Zygote**

TOPIC NO.2

2. The division of embryo into …… **three Germ Layers**

TOPIC NO.3

3. -----the study of birth defect……..… **Teratology**
4. There Are __approaches of anatomy……4

TOPIC NO.4

5. A condition in which cell lack the instructions for development…… **naïve**

TOPIC NO.5

6. Strategies of commitment……..2

TOPIC NO.6

7. Mode of commitment……..… **3**

TOPIC NO.8

8. Conditional specification gives rise to a pattern of embryogenesis……..… **regulative development.**
9. Regulative development is seen in most vertebrate embryos, and it is obviously- _____ of identical twins……..… **critical in the development**
10. The ability of the embryonic cell to change their fates to compensate for the missing part called ……..… **Regulation**

TOPIC NO.9

11. Insects such as_____of specification to commit their cells to particular fates…… **Drosophila use all three modes**

TOPIC NO.10

12. Congenital anomalies can be caused by…… **(both) genetic factors, environmental agents**

13. Which dermis give rise to brain…….. **Ectoderm.**

TOPIC NO.11

14. Cleavage sphere known as a…… **blastula**

TOPIC NO.12

15. In amphibians, metamorphosis is initiated by hormones from the tadpole's…… **thyroid gland**

16. The speed of metamorphosis is directly linked with…… **environment pressures**

TOPIC NO.13

17. Rana pipiens usually lay around_____eggs, while the bullfrog, Rana catesbiana, can lay…… **2500 & 20,000.**

TOPIC NO.14

18. Gastrulation in the frog begins at a point on the embryo surface roughly…….. **180 degrees**

TOPIC NO.15

19. ------exchanged nuclei b/w two morphologically distinct species…… **J.Hammerling**

20. Development us controlled at both the …… **transcriptional and translational level**

TOPIC NO.17

21. The individuals of each species, however is divided into two mating types Plus and minus…… **Chlamydomonas**

22. Chlamydomonas Which is usually haploid, having just ……**…one copy of each chromosome.**

TOPIC NO# 19-20

23. Major classes of cadherin include……., **N-cadherin**

24. Protocadherins have been found to be very important in separating the notochord from the other mesodermal tissues during ……..**Xenopus gastrulation**

TOPIC NO.25

25.Nucleosome is composed of an octamer of histone proteins wrapped with two loops containing approximately_____of DNA…… **140 base pairs**

TOPIC NO.26

26.Allows the mRNA to exit the nucleus…… **poly(A) tail**

TOPIC NO.27-28

27.Promoter is enhancer which is thought to …… **regulate transcription**

TOPIC NO.29-30-31

28.Pdx1 is specific for…….. **Pancreatic region of the endoderm.**

TOPIC NO.34-35

29.The first evidence suggesting the importance of sperm in reproduction came from a series of experiments performed by…… **Lazzaro Spallanzani**

TOPIC NO.38

30.It is estimated that the eggs of sea urchins contain----- different types of mRNA……**25,000 to 50,000**

31.The volume of a sea urchin egg is about…… **200 picoliters**

32.Protein synthesis is accomplished by ribosomes and……. **tRNA, which exist in the egg.**

TOPIC NO.40

33.During fertilization, these actin molecules polymerize to form long cables of actin known as…… **microfilaments**

34.During fertilization, these_____ molecules polymerize to form long cables of actin known as microfilaments……. **actin**

35.Vitelline envelop contains at least eight different…… **glycoproteins.**

36.The mammalian egg is also surrounded by a layer of cells called the ……..**,cumulus**.

TOPIC NO.48

37.Using immunological techniques, _____demonstrated that bindin is located specifically on the acrosomal process exactly where it should be for sperm-egg recagnition…… **Moy and Vacquier**

38.The acrosomal protein mediating this recognition is called bindin. In……..vacquier and co-workers isolated this nonsoluble 30.500-Ds (Dalton) protein from the acrosome of Strongylacentrotus purpuratus and found it to be capable of binding to dejellied eggs of the same species…… **1977**

TOPIC NO.47

39. Bleil and Wassarman (_____) isolated an 83-kDa glycoprotein, ZP3, from the mouse zona that was the active site for sperm binding. The other two zona glycoproteins they found, ZP1 and ZP2, failed to compete for sperm binding…… **1988**
40. The ____ is the specific glycoprotein in the mouse zona pellucida to which the sperm bind. …… **ZP3**
41. The _____ is a glycoprotein matrix, which is synthesized and secreted by the growing oocyte…… **Zona pellucida**

TOPIC NO.48

42. The difference between the acrosomal reaction in sea urchins and mammals may be due to the thickness of the extracellular envelopes surrounding the egg. In the sea urchin, the vitelline envelope is very thin and porous…… **True**
43. It appears that the acrosome-reacted sperm transfer their binding from ZP3 to the adjacent _____ molecules. After a mouse sperm has entered the egg, the egg cortical granules release their contents…… **ZP2**

TOPIC NO.49-50

44. The set of physiological changes that allow the sperm to be competent to fertilize the egg is called capacitation. The requirement for capacitation varies from species to species…… **True**
45. The Wilcox and colleagues_____ found that nearly all human pregnancies result from sexual intercourse during a 6-day period ending on the day of ovulation?…… **1995**
46. The molecular changes that account for capacitation are of _____types?…… **4**
47. In capacitation, the ……. the fluidity of the sperm plasma membrane is altered by the removal of cholesterol by albumin proteins found in the female reproductive tract? …… **First**
48. In capacitation, the _____, particular proteins or carbohydrates on the sperm surface are lost during capacitation…… **Second**
49. In capacitation, the _____, the membrane potential of the sperm becomes more negative as potassium ions leave the sperm…… **Third**
50. In capacitation, the _____, protein phosphorylation occurs…… **Fourth**

TOPIC NO.51-52

51. Recognition of sperm by the vitelline envelope or zona pellucida is followed by the lysis of that portion of the envelope or zona in the region of the sperm head by the enzymes…….… **Acrosomal**

52. The sperm and egg plasma membranes then join together, and material from the sperm membrane can later be found on the egg membrane. The sperm nucleus and tail pass through the resulting cytoplasmic bridge, which is widened by the …….… **Actin polymerization**

53. In mammals, the _____ proteins in the sperm plasma membrane are essential for sperm membrane-egg membrane fusion…….… **Fertilin**

54. The Ferilin _____ the sperm to the egg by binding to the 6β1 integrin protein on the egg plasma membrane. Moreover, like sea urchin bindin (to which it is not structurally related), fertilin has a hydrophobic region that could potentially mediate the union of the two membranes.…….… **Adheres**

55. In the sea urchin, all regions of the egg plasma membrane are capable of fusing with sperm. In several other species, certain regions of the membrane are specialized for sperm recognition and fusion. Fusion is an active process, often mediated by specific "fusogenic proteins? …….… **True**

56. Mouse fertilin is localized to the posterior plasma membrane of the sperm …….**head**

57.

TOPIC NO.53-54

58. The sea urchin egg has ___ mechanisms to avoid polyspermy?…….… **2**

59. The entrance of multiple sperm polyspermy leads to disastrous consequences in most organisms. In the sea urchin, fertilization by two sperm results in a _____ nucleus, in which each chromosome is represented three times rather than twice …….… **Triploid**

60. In normal _____, in which only one sperm enters the egg, a haploid sperm nucleus and a haploid egg nucleus combine to form the diploid nucleus of the fertilized egg (zygote). The centriole, which is provided by the sperm, will divide to form the two poles of the mitotic spindle during cleavage…….… **Monospermy**

TOPIC NO.55-56

61. The Fast block to polyspermy is achieved by changing the electric potential of the egg plasma membrane. This membrane provides a selective barrier between the egg cytoplasm and the outside environment, and the ionic concentration of the egg differs greatly from that of its surroundings…….… **True**

62. If we insert an electrode into an egg and place a second electrode outside it, we can measure the constant difference in charge across the egg …… **Plasma Membrane**

63. A third protein released by the cortical granules, a ___, hardens the fertilization envelope by crosslinking tyrosine residues on adjacent proteins?……**Peroxidase enzyme**

64. The fast sperm inhibition condition is maintained by the plasma membrane, which steadfastly inhibits the entry of…….ions into the oocyte and prevents potassium ions from leaking out into the environment……**Sodium**

TOPIC NO.57-58

65. Directly beneath the sea urchin egg plasma membrane is about…cortical granules……**15,000**

66. The removal of sperm is accomplished by the cortical granule reaction, a slower, mechanical block to _____ that becomes active about a minute after the first successful sperm-egg attachment…… **Polyspermy**

TOPIC NO.59

67. The _____ released by the cortical granules produce an osmotic gradient that causes water to rush into the space between the plasma membrane and the vitelline envelope, causing the envelope to expand and become the fertilization envelope…… **Mucopolysaccharides**

68. Finally, a fourth cortical granule protein, _____, forms a coating around the egg…… **Hyalin**

69. The egg's microvilli attach to…….provides support for the blastomeres ……**hyaline layer.**

TOPIC NO.60-61

70. The release of calcium from intracellular storage can be monitored visually using calcium-activated luminescent dyes such as……**fura-2**

71. Release of Ca from intracellular storage controlled by-----dye…… **Aequorin**

72. The drug is …….. calcium ionophore (a compound that transports free calcium ions across lipid membranes)……**A23187**

TOPIC NO.62-63

73. Slow block, initiated by the intracellular release of……**calcium ions**

74. The fast block, initiated by---------- into the cell in sea urchin…… **sodium influx**

TOPIC NO.64-65

75. Which has two parts in cell cycle options ……**Blastomere**

76. Transition from fertilization to cleavage is caused by factor……**MPE**

TOPIC NO.66-67

77. The mitotic division of the nucleus is called…… **karyokinesis**

78. The large subunit is called ……**cyclin B.**

TOPIC NO.69

79. Egg activates due to which ion more……**Ca**

TOPIC NO.70

80. Eggs having sparse and equally yolk space are called……**isolecithal**

81. Telolecithal is cleavage is found in eggs of……**Birds and Fishes**

82. The eggs of insects have their yolk in the center……**centrolecithal**

83. Divisions of the cytoplasm occur only in the rim of cytoplasm around the periphery of the cell……**superficial cleavage**

TOPIC NO#71-72

84. Sea urchins exhibit which cleavage…… **radial holoblastic cleavage.**

TOPIC NO.73-74

85. Archenteron is used for …… **primitive gut**

86. Inner surface cells thin 30 x 5um process……..**Filopodia**

87. These cells begin extending and contracting long, thin (30 × 5 μm) processes called…… **filopodia**

88. Cells, derived from the micromeres, are called the……**Primary Mesenchyme.**

TOPIC NO.75-76

89. Spiral cleavage found in ……**Snails**

90. _____is characteristic of several animals'groups, including annelid worms, some flatworms, and most mollusks (snails)…… **Spiral Holoblastic Cleavage**

91. Spiral holoblastic is characteristic of several animals 'groups, including_____, some flatworms, and most mollusks (snails)…… **annelid worms**

92. Blastulae produced by radial cleavage have no blastocoel and are called…… **Stereoblastulae**

93. Mollusca follow which types of cleavage…… **Spiral Holoblastic Cleavage**

94. The orientation of the cleavage plane in snail eggs controlled by……**Cytoplasmic factor**

TOPIC NO.77-78

95. Shell formation in ……..**snail**

96. The snail stereo blastula is relatively…… **Small**

97. While gastrulating process in snail the micromeres will cover the entire……**Embryo**

98. Gastrulation in Snails is accomplished primarily by……**Epiboly**

99. The resulting larvae of snails is called _ which resemble tiny bear traps ……**Glochidia**

TOPIC NO.79-80

100. Ascidian, members of the phylum……**Tunicate**

101. Ascidians are the ……**Invertebrates**

102. Ascidians have …… **notochord**

103. They have a notochord as larvae but they lack ……**Vertebrae**

104. It secretes a_____tunic (which gave the name "tunicates" to these creatures) ……**Cellulose**

105. In Tunicate the second cleavage……**Meridional cleavage**

106. In Tunicate cleavage during the next three divisions, differences in cell size and shape highlight the _____ of these embryos…… **Bilateral Symmetry**

107. In Tunicate the second cleavage creates two large interiors blastomeres……**A and D**

108. In tunicate the cleavage at 32-cell stage, a small blastocoel is formed and_ begins ……**Gastrulation**

TOPIC NO.81-82

109. Tunicate gastrulation is characterized by the invagination of the endoderm. the involution of the mesoderm and the epiboly of the……**Ectoderm**

110. Tunicate gastrulation is characterized by the invagination of the ……**Endoderm**

111. The second step of gastrulation involves the…….. **Involution of The Mesoderm**

112. Pairs of blastomeres appear to lead this invagination into the center of the embryo. This invagination forms a blastopore whose lips will become the……**Mesodermal cells**

113. The _____ that are the precursors of the neural tube invaginate into the embryo and are enclosed by neural folds …… **Dorsal Ectodermal Cells**

114. While gastrulation of tunicate the presumptive tail muscle cell ____ are on the posterior lip of the blastopore……**Yellow crescent**

115. The lateral lips of the blastopore comprise these cells that will become……**Mesenchyme**

116. The second step of gastrulation in Tunicates involves the involution of the……**Mesoderm**

117. The dorsal ectodermal cells that are the precursors of the neural tube invaginate into the embryo and are enclosed by neural folds. This forms the neural tube, which will form a ____ anteriorly and _____ posteriorly……**Brain and Spinal chord**

TOPIC NO.83-84

118. Cell fates are determined by signaling. The micromeres constitute a major signaling center. β-catenin is important for the inducing capacity of the …… **Micromeres**

119. --------constitute a major signaling center…….. **Micromeres**

120. A mammalian blastula is called a……. **Blastocyst.**

121. A_____ is an embryonic structure composed of blastomeres ……**Blastula**

122. The blastomere cell cycle is governed by the synthesis and degradation of ……. **Cyclin.**

123. A _____is a cell derived from cleavage in an early embryo……**Blastomere**

124. In Nematode cleavage is ……**Rotational**

125. The movements of gastrulation include ……. **invagination, involution, ingression, delamination, and epiboly**

126. Cyclin synthesis promotes the formation of_____ **MPF**

127. Cyclin synthesis promotes the formation of MPF, and MPF promotes……**Mitosis**

TOPIC NO.85-86

128. The cytoplasm of the blastodisc becomes the embryo, this type of meroblastic cleavage is called……**Discoidal**

129. The first------- divisions occur synchronously, forming a mound of cells that sits at the animal pole of a large yolk cell……..**12**

130. only the cytoplasm of the blastodisc becomes the embryo, this type of meroblastic cleavage is called discoidal present in _____ **fishes**

131. The yolk syncytial layer is formed at the____ when the cells at the vegetal edge of the blastoderm fuse with the underlying yolk cell……**9th cell cycle**

TOPIC NO.87-88

132. The resulting larvae of snail is called ____which resemble tiny bear traps……**Glochidia**

133. The first cell movement of fish gastrulation is the_____of the blastoderm cells over the yolk…….**Epiboly**

TOPIC NO.89

134. The cells adjacent to the chordamesoderm, the paraxial _____ cells, are the precursors of the mesodermal somite…… **Mesoderm**

135. The cells of both the epiblast and hypoblast intercalate on the future dorsal side of the embryo to form a localized thickening, the ……. **Embryonic shield**

136. After the ___- cells have covered about half the zebra fish yolk cell, a thickening occurs throughout the margin of the epibolizing blastoderm……**Blastoderm**

137. After the blastoderm cells have covered about half the zebra fish yolk cell, a thickening occurs throughout the margin of the epilogizing blastoderm. This thickening is called the……. **Germ Ring**

138. The cells adjacent to the chordamesoderm, the paraxial mesoderm cells, are the precursors of the _____ in fish ……**Mesodermal somite**

139. The neural form……..**Keel Fish**

TOPIC NO.90-91

140. Fertilization of the chick egg occurs in the……**Oviduct**

141. The first cleavage furrow appears centrally in the……. **Blastodisc**

142. In birds egg cleavage occurs only in the blastodisc, a small disc of cytoplasm _ in diameter at the animal pole of the egg cell……. **2-3 mm**

143. The peripheral ring of blastoderm cells that have not shed their deep cells constitutes the …….. **area opaca**

144. Between the area pellucida and the area opaca thin layer of cells called the ……. **marginal belt or marginal zone**

145. Between the blastoderm and the yolk is a space called the……**Sub-Germinal Cavity**

TOPIC NO. 92-93

146. The center of this node contains a funnel-shaped depression through which cells can pass into the blastocoels called as…… **Primitive pit**

147. The major structural characteristic of avian, reptilian, and mammalian gastrulation is the……..**Primitive Streak.**

148. The depression is called the……..**Primitive Groove**

149. This streak is first visible in birds as a thickening of the _____ at the posterior region of the embryo, just anterior to Koller's sickle……**Epiblast**

150. At the anterior end of the primitive streak is a regional thickening of cells called the…….**Primitive knot or Hensen's node**

TOPIC NO.94-96

151. The mammalian oocyte is released from the ovary and swept by the fimbriae into……..**The Oviduct.**

152. Fertilization occurs in the_____of the oviduct, a region close to the ovary……**Ampulla**

153. Fertilization occurs in the ampulla of the…….**oviduct**

154. The most crucial, difference between mammalian cleavage and all other types involves the phenomenon of……**Compaction**

155. In mammalian cleavage at 64-cell stage, the inner cell mass and the _____ cells have become separate cell layers, neither contributing cells to the other group…….**Trophoblast**

156. In mammalian cleavage _____ catches the blastocyst on an extracellular matrix containing collagen, laminin, fibronectin, hyaluronic acid, and heparin sulfate receptors……..**Endometrium**

157. During mammalian cleavage the trophoblast cells secrete fluid into the morula to create a blastocoels this process called as……..**Cavitation**

TOPIC NO.97-100

158. The mammalian embryo obtains nutrients directly from its ……**Mother**

159. Chorion is derived from _____ cells ……. **trophoblast**

160. This fetal organ the_____is derived primarily from embryonic trophoblast cells……**Chorine**

161. The fetal portion of placenta formed by …… **Chorion**

162. Multinucleated type of cell form-……..**syncytiotrophoblast**

163. Mother provides fetus with…….**Nutrients and Oxygen**

164. It will induce the uterine cells to form the maternal portion of the placenta, the…….**decidua**

165. While gastrulation process in mammalian _____ becomes rich in the blood vessels that will provide oxygen and nutrients to the embryo……. **Decidua**

166. The decidua becomes rich in the blood vessels that will provide ___ and nutrients to the embryo. …….**Oxygen**

167. The cytotrophoblast initially adheres to the ___ through a series of adhesion molecules……. **Endometri**

168. The narrow connecting stalk of extra-embryonic _____ that links the embryo to the trophoblast eventually forms the vessels of the umbilical cord …….**Mesoderm**

169. The first segregation in mammalian cells within the inner cell mass results in the formation of the…….**hypoblast (sometimes called the Primitive endoderm)**

170. The original type of trophoblast cells constitutes a layer called the……**Cytotrophoblast**

171. Mother provides the fetus with nutrients and oxygen, and the fetus sends its waste products _____ into the maternal circulation……..**Carbon dioxide and urea**

TOPIC NO.101-104

172. Fishes,_____, and birds undergo discoidal meroblastic cleavage, wherein the early cell divisions do not cut through the yolk of the egg …….**Reptiles**

173. Discoidal meroblastic cleavage occur in……. **birds and reptiles**

174. The prechordal _____ induces the formation of the forebrain; the chordamesoderm induces the formation of the midbrain, hindbrain, and spinal cord……. **Plate**

175. Mammals undergo holoblastic_____cleavage, …….**Rotational**

176. In birds,_____ is critical in determining the anterior-posterior axis……. **Gravity**

177. In birds_____ differences appear crucial for distinguishing dorsal from ventral ……. **pH**

178. Hox genes pattern the anterior-posterior axis and help to specify positions along that……. **Axis**

179. Which forms spinal cord, hind brain and mid brain …….**Chordamesoderm**

TOPIC NO.105-106

180. Each ICM (Inner Cell Mass) cell has the_____potency……. **Same**

181. Fraternal twins are the result of ____ separate fertilization events ……**Two**

182. The remaining Identical twins share a common……..**Chorion**

183. Identical twins occur in roughly -------- of human births. ……**0.25%**

184. About___of identical twins have two complete and separate chorions……. **33%**

TOPIC NO.107-108

185. The rudiment of the central nervous system, is called …….. **Neurulation**

186. Neurulation in _____ is exclusively secondary ……**Fishes**

187. Nervous system develops by……**Neural Tube**

188. Which type of neurulation in fish ……**Secondary**

189. In amphibians most of the xenopus most of tadpole is formed by_____and tail of neural tube formed by 2ndry neurulation……**Primary Neurulation**

TOPIC NO.109

190. In primary neurulation, the original ectoderm is divided into_____sets of cells……**Three**

TOPIC NO.110

191. The neural folds migrate toward the midline of the embryo, eventually fusing to form the_____beneath the overlying ectoderm……. **neural tube**

TOPIC NO.112

192. Failure to ___the human posterior neural tube regions at day 27……. **Close**

193. Failure to close the anterior neural tube regions results in a lethal condition…….. **Anencephaly.**

TOPIC NO.113-114

194. In frogs and chicks_____is usually seen in the neural tube of the lumbar (abdominal) and tail vertebrae…….**secondary neurulation**

195. The growing region at the tip of the lip is called the……**Chord Neural Hinge**

196. Secondary neurulation involves the making of a _____ and its subsequent hollowing into a neural tube. ……. **Medullary Cord**

TOPIC NO.115-118

197. Hindbrain…… **Rhombencephalon**

198. The Myelencephalon eventually becomes the……. **Medulla oblongata**

199. One agent of ventral specification is the _____hedgehog protein…… **Sonic**

200. Sonic hedgehog is a type of……**Protein**

201. The telencephalon will eventually form the cerebral hemispheres, and the diencephalon will form the thalamic and _____brain regions……**Hypothalamic**

202. Retinoic Acid comes from …… **Somite's**

TOPIC NO.119-120

203. Mantle cells differentiate into…… **Gray Matter**

204. The mantle zone cells differentiate into both ……**Neurons and Glia**

TOPIC NO.121-122

205. Marginal zone to form clusters of neurons called…… **Nuclei**

206. Meanwhile, the original ependymal layer of the cerebellum generates a wide variety of neurons and glial cells, including the distinctive and large……. **Purkinje neurons.**

TOPIC NO.123

207. The human brain consists of…… **10^{11} neurons**

208. Human brain contains ……**10^{12} Glial Cells**

209. Those cells that remain integral components of the neural tube lining become____ cells……**Ependymal**

TOPIC NO.124

210. Dendrites are often numerous and do not extend far from the neuronal cell body, or____ axons may extend for several feet……**Soma**

TOPIC NO.125-126

211. The cells at dorsal most portion of neural tube become …….**Neural Crest Cells.**

212. The vagal (neck) neural crest lies opposite chick somites…… **1-7**

213. The neural crest cells originate at the____most region of the neural tube. ……**Dorsal**

214. Neural crest divides functionally indomains……**4**

TOPIC NO.127-130

215. ---------- plays a major role in eye formation……**Pax6**

216. Paracrine factors such as_____ and FGF7 are important in normal skin development. ……**TGF**

217. Paracrine factors such as TGF- and FGF7 are important in normal……**Skin Development.**

218. The cerebral cortex in humans has _____ layers…… **Six**

219. TGF-β family secreted from the surface_____and roof of the neural tube. ……**Ectoderm**

220. The space between two neurons which transmit are released……..**Synapse**

TOPIC NO.131-132

221. Paracrine factors instruct the_____to become muscles by inducing them to synthesize the Myoid protein……..**Myotome Cells**

222. Paracrine factor instruct myotome cell to become…… **Muscles**

223. Myogenic bHLH proteins are specific for development of…….**Muscle Cell**

TOPIC NO.133-134

224. The second step is the ____of the myoblasts together into chains……. **Alignment**

225. When myoblasts stop dividing, secrete _____**Fibronectin**

TOPIC NO.135-136

226. There are____major modes of bone formation, or osteogenesis ……**Two**

227. The direct conversion of mesenchymal tissue into bone is called ……..**Intramembranous Ossification**

TOPIC NO.137-138

228. The osteoblasts secrete a collagen-proteoglycan _____ that is able to bind calcium salts……**Matrix**

229. The entire region of calcified spicules becomes surrounded by _____mesenchymal cells that form the Periosteum ……**Compact**

TOPIC NO.139-142

230. Endochondral ossification involves the formation of ……..**Cartilage**

231. Mutations of the __ cause campomelic dysplasia, a rare disorder of skeletal development that results in deformities of most of the bones of the body……**SOX9 Gene**

232. Mutations of the SOX9 cause by…….. **campomelic dysplasia**

233. Mutations of the_____cause camptomelic dysplasia, a rare disorder of skeletal development that results in deformities of most of the bones of the body…….**SOX9 gene**

234. ____and Scleraxis are the transcription factors are thought to activate cartilage-specific genes……**pax1**

235. Scleraxis is expressed in the___from the sclerotome……**mesenchyme**

236. _____phase involves the invasion of the cartilage model by blood vessels ……**Fifth**

237. The SOX9 gene, which ____ a DNA-binding protein …… **Encodes**

TOPIC NO.143-144

238. Osteoclasts are probably derived from the ___ precursors as macrophage blood cells…… **Same**

BEST OF LUCK